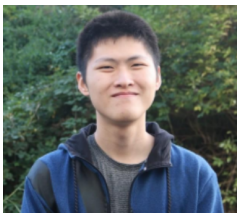
Curriculum Vitae



Personal Information

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Self-Intro

Physics Masters graduate with a strong understanding of semiconductor process technologies on top of an interdisciplinary background. Interned at Dutch public research institutions i.e. Advanced Research Center for Nanolithography (ARCNL) and AMOLF for applied research projects. Experienced with operating and calibrating complex vacuum systems and (coherent) optical setups. I am eager to kickstart my career in the semiconductor industry at Lam Research, with preference towards process support engineer roles.

Internships

ARCNL (in partnership with ASML) MSc Thesis, Full Time *Aug. 2023–Aug. 2024*
Materials & Surface Science for EUVL group, Supervised by Dr. Roland Bliem

Investigated possible oxidation-based damage mechanisms caused by pulsed Deep Ultraviolet (DUV) exposure to metallic surfaces. Found DUV illumination to have negligible impact on thermal-driven oxidation.

- **Took initiative of operating and maintaining lab equipment.** I independently operated a metal deposition (physical vapor deposition) setup, DUV laser beam line and X-ray based characterization technique under compliance; running pre-use checks, calibration and maintenance with documentation independently.
- **Designed my own project and experiments.** I utilized Design-of-Experiments (DoE) principles to create a timeline of milestones and goals that ensures completion within the time limit set by education requirements. Designed my own experiments and data analysis processes and establish a virtuous cycle of continuous improvement.
- **Championed improvements on troubleshooting routines.** I provided technical support and aligned with technicians during troubleshooting and safety procedure updates to find short-term fixes and long-term solutions for the aforementioned complex systems.
- **Organized inter-departmental communication efforts.** I organized and chaired a departmental meeting (approx. 20 attendees in terms of size) for relevant stakeholders to update one another on new developments and findings.
- **Presented my research to industry and academic experts.** I reported technical findings to academic and ASML collaborators from the Wafer Damage Control Project and the UvA Quantum Materials group.

AMOLF (part of the Dutch Research Council) BSc Thesis, Part-Time *Jan. 2021–June 2021*
Hybrid Solar Cells group, Supervised by Dr. Bruno Ehrler and Dr. F.R. Bradbury

Tested the accuracy of an Machine Learning Model built for locating optimal solar cell placement. I found the model have an accuracy of $< 50\%$, demonstrating a need for further improvements.

- **Creatively solve unforeseen problems**. I was able to gather experimental data outdoors despite COVID restrictions by developing an optical setup with procedures compliant to health guidelines.
- **Reported my progress and incorporate feedback from others**. I reported preliminary results to colleagues and superiors during weekly update meetings and incorporated constructive feedback coming from their end.
- **Automated testing protocols and visualization.** I wrote scripts with comprehensive documentation designed for easy access and retrieval of results.

Education History

MSc Physics & Astronomy University of Amsterdam *Sept. 2021–Aug. 2024*
Advanced Matter and Energy Physics (AMEP) track
Relevant coursework :
Condensed Matter I, Nanophotonics, Programming for AMEP (Python & Mathematica), Quantum Optics, Surface and Interface Science, The Science and Technology of Nanolithography

BSc Liberal Arts & Sciences Amsterdam University College *Sept. 2018–Aug. 2021*
Physics track
Relevant coursework :
Numerical Methods (MATLAB), Probability and Statistics (R), Physics Lab

Other Experiences

AMEP Seminar Member University of Amsterdam *Sept. 2021–July 2022*

Assisted in organizing monthly Amsterdam-based physics seminars (target audience being UvA students). Took meeting notes and ensured accessibility of all records. Acted as liaison towards university faculty to help reach a balance between seminar attendee satisfaction and budgetary realities. Drafted sample letters and interaction protocols to educate future members.

Part-time staff of SCES RAI Amsterdam *July 2022*

Assisted with seminar operations and support attendees at the International Conference on Strongly Correlated Electron Systems (SCES). Ensured seminars conclude within designated time-slots, acted as first-of-line technical support, resolve miscellaneous issues for attendees e.g. poster presentation.

Co-Founder of Gaze Amsterdam University College *2020-2021*

Co-founded a committee for organizing student-focused monthly cultural events. Communicated and Promoted our cause to ensure adequate financial support from the student body and association.

Member of 3D Printing club Amsterdam University College *2019-2021*

Actively participated in building and maintaining 3D Printers. Maintained an archive of meetings, past maintenance logs and project documentation. Liaison-ed with university faculty for funding and balance capabilities with budgetary limitations.

Co-Presenter at 22nd IIS Netherhall house, London *July 2020*

Co-presented *The Multiverse versus the Copenhagen interpretation of Quantum Mechanics* at the 22nd International Interdisciplinary Seminar. Presented a highly technical subject to researchers and students (approx. 20-30 attendees) from technical & non-technical backgrounds alike.

Skills

Soft Skills : Communicative, Detail-Oriented, Motivated, Independent

Languages : English (Fluent), Chinese (Fluent), Dutch (Limited), German (Elementary)

Software & Programming : Python, MATLAB, Mathematica, R

Technical Familiarities : (Coherent) Optical systems, Ultra-High Vacuum, Thin Film Deposition & Characterization

References

— Dr. Roland Bliem (MSc Thesis Supervisor) : R.Bliem@arcnl.nl
— Dr. Paul Planken (MSc Lecturer & Thesis Reader) : P.Planken@arcnl.nl
— Dr. Bruno Ehrler (BSc Thesis Supervisor) : B.Ehrler@amolf.nl

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To whom it may concern,

As a recent graduate from University of Amsterdam (UvA) with a Master's in Physics, I am eager to apply my Physics education, communication skills, and focus on growth to engineering solutions for semiconductor manufacturing difficulties. I am thus interested in kick-starting my engineering career by joining your company.

During my studies in Physics, I developed skills that I believe directly align with this position, including:

1. **Semiconductor Affinity** – During my studies I focused on technologies and principles behind semiconductor manufacturing. Courses such as *Quantum Optics*, *Surface & Interface Science* and *The Science & Technology of Nanolithography* built my understanding on **coherent optics**, **wafer metrology**,, **thin film properties & dynamics** and **ultraviolet photolithography**. Upon this theoretical foundation I built up practice experience by interning full-time for a year at the Advanced Research Center for NanoLithography (ARCNL), a research institution founded under a public-private partnership between the Dutch government and ASML, under Dr. Roland Bliem. There I investigated oxidation-based damage mechanisms of electronic materials e.g. copper caused by exposure to **Deep Ultraviolet** (DUV) pulses. In collaboration with academic and ASML experts, I developed an intuition for working with **Ultra-High Vacuum** systems and DUV beamlines.
2. **Communication is Key** – As a Physicist, I've collaborated with peers and stakeholders within and outside my field. During my ARCNL internship, participating in horizontal knowledge transfers is crucial. At the research group level, attending progress meetings and providing user feedback during troubleshooting sessions is part of my routine. At the departmental level, I chaired and organized a meeting for members to announce new projects and updates of ongoing ones. Outside the institute, my results are presented to various groups, such as the Quantum Materials group from the UvA Institute of Physics, and the Wafer Damage Control Project in collaboration with Dutch academic and industry experts.
3. **Motivated Learner** – I believe in life-long learning and am a fast learner for unfamiliar technologies. To pursue a world-class physics education, I left my home country and studied in the Netherlands for 6 years, ultimately completing a Physics Masters with an interdisciplinary background. I learned the ins and outs of complex systems e.g. **Physical Vapor Deposition** & **X-ray Photoelectron Spectroscopy** within a few months and operate them independently, demonstrating the capacity to learn and adapt to new technologies. Furthermore I learned how to calibrate and conduct light maintenance for these machines, an experience that translates well into an engineering role.

I am particularly drawn to your company because of the room for professional growth and my interest in the tech. During my studies and ARCNL internship, I learned a great deal about semiconductor processes, its challenges, and the technical know-how to address them. Collaborations between ARCNL and ASML, such as the Wafer Damage Control Project with Dr. Vina Faramarzi, gave me a good impression about the industry with the availability of technical opportunities and its critical position within the modern digital landscape. After my studies, I decided to pursue a technical career with interpersonal opportunities in this field. I particularly like how such opportunities offers the chance to interact and learn the needs and wants of different stakeholders. One would spend plenty of time translating customer needs into technical solutions that are achievable with existing manufacturing capabilities. I look forward to hone my technical competencies while furthering communication skills.

I am confident that my familiarity with nanolithographic technologies, communicative affinity and commitment to growth will make me a valuable addition to your team. I would appreciate the chance to discuss how my background and passion align with the goals of your company.

Kind Regards,

Max Wu